D3: Discussion #3 – Item #1

Data Visualization in Python: U.S. Vice Presidents by Age | Michelle Feliciano | September 2026

Built in Python (pandas, matplotlib, seaborn) — workflow shown below as it would run in Google Colab, with each code cell followed by its output.

Data source: Wikipedia, “List of vice presidents of the United States by age,” https://en.wikipedia.org/wiki/List_of_vice_presidents_of_the_United_States_by_age (birth dates, term dates; all 50 vice presidencies, John Adams through J.D. Vance).

Step 1 – Load the data and compute age at start of term

[ ]:

```
import pandas as pd

df = pd.read_csv("vp_ages.csv",
                 parse_dates=["birth_date", "death_date",
                              "term_start", "term_end"])

def age_in_years(start, end):
    return (end - start).days / 365.25

df["age_at_start"]   = df.apply(
    lambda r: age_in_years(r["birth_date"], r["term_start"]), axis=1)
df["term_start_year"] = df["term_start"].dt.year

print(df[["number", "name", "term_start_year", "age_at_start"]]
      .round(1).head(6).to_string(index=False))
print(df["age_at_start"].describe().round(1))
```

```
number               name  term_start_year  age_at_start
     1         John Adams             1789          53.5
     2   Thomas Jefferson             1797          53.9
     3        Aaron Burr              1801          45.1
     4     George Clinton             1805          65.6
     5      Elbridge Gerry             1813          68.6
     6 Daniel D. Tompkins             1817          42.7
                      ...

count    50.0     mean   54.6     std    8.5
min      36.1     25%    49.4     50%    53.6
75%      60.3     max    71.2
```

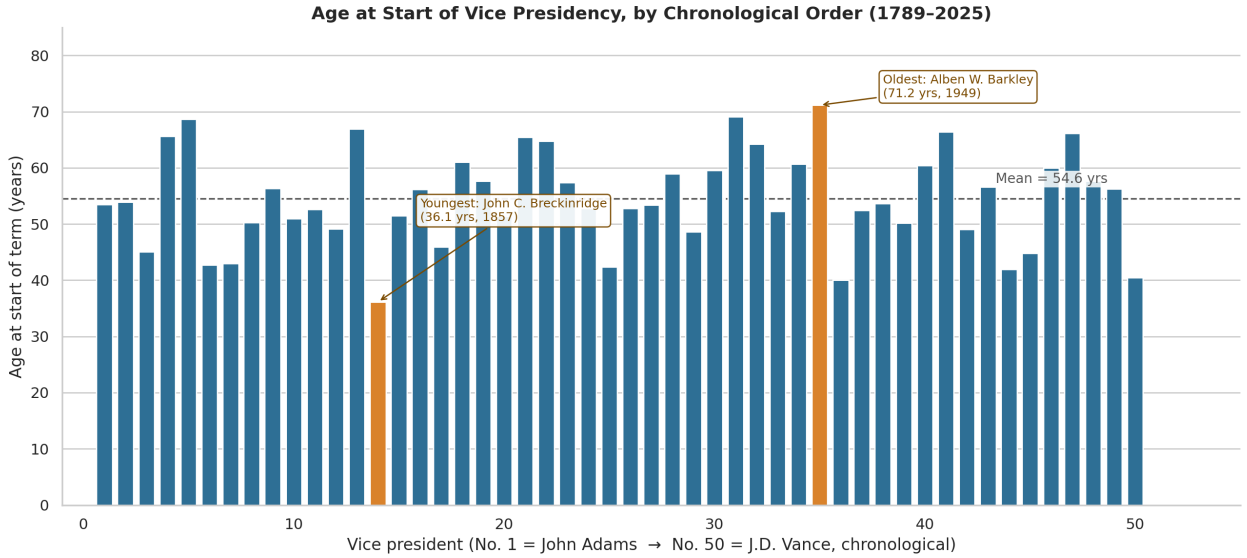
Step 2 – Visualization 1: bar chart of age at start of term

[ ]:

```
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style="whitegrid")

colors_ = ["#2E6F95"] * len(df)
colors_[df["age_at_start"].idxmin()] = "#D9822B"   # youngest
colors_[df["age_at_start"].idxmax()] = "#D9822B"   # oldest

fig, ax = plt.subplots(figsize=(13, 6))
ax.bar(df["number"], df["age_at_start"], color=colors_, width=0.75)
ax.axhline(df["age_at_start"].mean(), color="#555", ls="--", lw=1.2)
ax.set_xlabel("Vice president (No. 1 = John Adams -> No. 50 = J.D. Vance)")
ax.set_ylabel("Age at start of term (years)")
ax.set_title("Age at Start of Vice Presidency, by Chronological Order (1789-2025)")
plt.show()
```

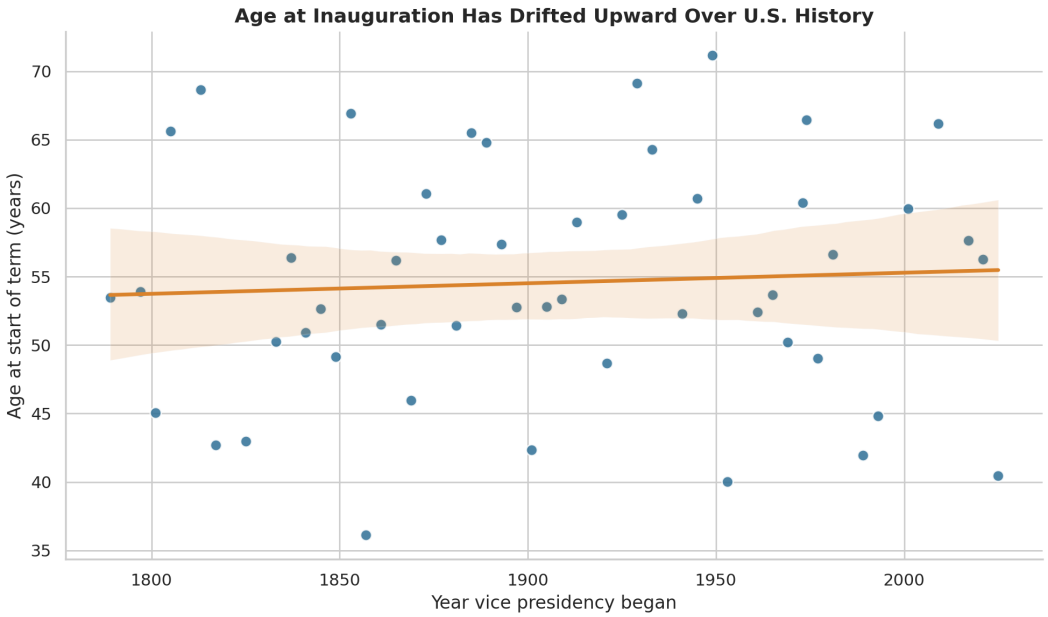


Reading this chart: the dashed line marks the average age (54.6 years) at which a vice president has taken office. John C. Breckinridge (1857) was the youngest at 36.1 years; Alben W. Barkley (1949) was the oldest at 71.2 years — a 35-year spread across the same office.

Step 3 – Visualization 2: has age at inauguration changed over time?

[ ]:

```
fig, ax = plt.subplots(figsize=(10, 6))
sns.regplot(
    data=df, x="term_start_year", y="age_at_start", ax=ax,
    scatter_kws=dict(color="#2E6F95", s=55, alpha=0.85, edgecolor="white"),
    line_kws=dict(color="#D9822B", linewidth=2.5), ci=95,
)
ax.set_xlabel("Year vice presidency began")
ax.set_ylabel("Age at start of term (years)")
ax.set_title("Age at Inauguration Has Drifted Upward Over U.S. History")
plt.show()
```



Reading this chart: each point is one vice presidency plotted by the year it began. The trend line (with a 95% confidence band) has a slight positive slope, but the correlation is weak (r = 0.06) — age at inauguration has stayed noisy across 236 years rather than climbing steadily.